

INTERNSHIP PROJECT REPORT ON

“PRODUCT DATA MANAGEMENT SYSTEM”

A report on the project developed during internship at

Dhruv Compusoft Consultancy Pvt Ltd

Submitted by

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CERTIFICATE

This is to certify that Sanjana A C, working as an Intern at Dhruv Compusoft Consultancy Pvt Ltd, has worked on the internship project entitled “Product Data Management System” under the guidance of Uday Shetty Kumar. The project is a web application developed using Java, Spring Boot, Spring MVC, Thymeleaf, HTML, CSS, Spring Data JPA, Lombok and PostgreSQL.

DECLARATION

I hereby declare that this report presents the work carried out by me during my internship at Dhruv Compusoft Consultancy Pvt Ltd. The report describes the development of the “Product Data Management System” and the work completed as part of my internship.

Signature: Sanjana A C

Date: 05/10/2026

ACKNOWLEDGEMENT

I would like to express my sincere thanks to Dhruv Compusoft Consultancy Pvt Ltd for giving me the opportunity to work as an Intern. I am grateful to my mentor, Uday Shetty Kumar, for his guidance and support throughout the development of the Product Data Management System. This internship helped me gain practical experience in Java, Spring Boot, Spring MVC, Thymeleaf, HTML, CSS, Spring Data JPA, Lombok and PostgreSQL.

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Chapter 1

PREAMBLE

1.1 INTRODUCTION

The Product Data Management System (PDMS) is a simple web application developed during my internship at Dhruv Compusoft Consultancy Pvt Ltd. It is used to manage product-related information such as parts, documents, manufacturers and manufacturer parts. The application provides a single place where a user can view and maintain important product details instead of managing them manually.

The project uses Java and Spring Boot for the application, Thymeleaf for the web pages, HTML and CSS for the user interface, Spring Data JPA for persistence, Lombok for constructor generation, and PostgreSQL for storing data.

1.2 STATEMENT OF PROBLEM

Managing product information manually in separate files can become difficult when the number of records increases. Searching for information, maintaining related records and keeping product details organized can take extra time. A simple web-based system can make these tasks easier.

1.3 OBJECTIVE OF THE PROBLEM

- To create a simple system for managing product data.
- To manage part information in one application.
- To manage document information and versions.
- To manage manufacturer information.
- To manage manufacturer part information and linked parts.
- To provide a login page for users.
- To show useful totals on the dashboard.
- To store application data in PostgreSQL.

Chapter 2

LITERATURE SURVEY

Product data management systems are commonly used to store and organize product information. Modern web applications allow users to access such information through a browser instead of working directly with separate files.

Spring Boot is widely used for developing Java web applications because it reduces configuration and provides a structured application framework. Spring MVC provides controllers for handling requests, while Thymeleaf can display Java data inside HTML pages.

Spring Data JPA provides repository interfaces for database operations. It allows Java entity classes to be stored and retrieved from a relational database without writing every database operation manually.

PostgreSQL is a relational database management system suitable for structured application data. In this project, it is used to persist parts, documents, manufacturers and manufacturer-part records.

The project combines these technologies into one simple application containing login, dashboard, part management, document management, manufacturer management, manufacturer-part management and consolidated data viewing.

Chapter 3

SYSTEM REQUIREMENTS SPECIFICATION

3.1 HARDWARE REQUIREMENTS

- Computer or laptop
- Minimum 4 GB RAM recommended
- At least 2 GB free space for Java, Maven and project files
- Keyboard and mouse
- Monitor with normal display resolution

3.2 SOFTWARE REQUIREMENTS

- Operating System: Windows / Linux / macOS
- Programming Language: Java 25
- Framework: Spring Boot 3.5.6
- Web Technology: Spring MVC and Thymeleaf
- Frontend: HTML and CSS
- Persistence: Spring Data JPA / Hibernate
- Database: PostgreSQL
- Database Driver: PostgreSQL JDBC
- Build Tool: Maven
- Additional Library: Lombok
- Browser: Chrome, Edge or Firefox
- IDE: Eclipse, IntelliJ IDEA or Spring Tool Suite

Chapter 4

DESIGN

4.1 EXISTING SYSTEM

In a simple manual system, product and manufacturer information may be maintained in separate spreadsheets or files. The user has to search through records and update information manually. Keeping related part, document and manufacturer information together may also require additional work.

4.2 PROPOSED SYSTEM

The proposed Product Data Management System provides a single web application for managing parts, documents, manufacturers and manufacturer parts. The user can log in and use separate pages for each type of information.

- Simple browser-based interface
- Login and logout
- Create, view and delete records
- Dashboard with totals
- Consolidated search/data viewing page
- PostgreSQL-based data storage
- Layered controller, service and repository structure

4.3 LOW LEVEL DESIGN

The application is divided into small parts so that each part has a clear purpose.

Part	Purpose
Controller	Receives requests from the web pages and sends responses
Service	Performs the main application operations
Model	Stores the structure of Part, Document, Manufacturer, ManufacturerPart and User data
Repository	Uses Spring Data JPA to access PostgreSQL
Thymeleaf Pages	Display forms, lists and dashboard information
CSS	Controls the appearance of the pages

Basic flow:

User → Web Page → Controller → Service → Repository → PostgreSQL

Chapter 5

IMPLEMENTATION

5.1 LOGIN AND SESSION

The LoginController receives the username and password. LoginService validates the supplied credentials. If the credentials are correct, the username is saved in the HTTP session and the user is redirected to the dashboard. Logout invalidates the session and returns the user to the login page.

5.2 PART MANAGEMENT

PartController and PartService are used for part operations. When a part is created, PartService generates the next numeric part number, creates a Part object and saves it through PartRepository. The parts page displays the stored records and provides a delete operation.

5.3 DOCUMENT MANAGEMENT

DocumentController and DocumentService are used to manage documents. A document contains a number, name, description and version. The service generates the next document number, saves the record through DocumentRepository and supports viewing and deletion.

5.4 MANUFACTURER MANAGEMENT

ManufacturerController and ManufacturerService manage manufacturer records. A manufacturer contains an identifier, name and country. ManufacturerRepository is used to store and retrieve the records.

5.5 MANUFACTURER PART MANAGEMENT

ManufacturerPartController and ManufacturerPartService manage manufacturer-part records. A record contains a manufacturer part number, manufacturer name and linked part number. The service generates the next numeric identifier and stores the record through ManufacturerPartRepository.

Chapter 5

IMPLEMENTATION

5.6 DASHBOARD

DashboardController collects totals from the different services and sends them to the dashboard page. The dashboard displays total parts, total documents, total manufacturers and total manufacturer parts.

5.7 SEARCH

SearchController provides the /search page. The current implementation retrieves parts, documents, manufacturers and manufacturer parts and displays them together. The current page therefore acts as a consolidated data-viewing page rather than a text-filter form.

5.8 POSTGRESQL DATA HANDLING

The application is configured to connect to the PostgreSQL database named pdms on the local PostgreSQL server. Spring Data JPA and Hibernate map the Java entity classes to database tables. The application properties contain the JDBC connection and JPA configuration.

5.9 MAIN CONTROLLERS

Controller	Main Purpose
LoginController	Login and logout
DashboardController	Dashboard totals
PartController	Part operations
DocumentController	Document operations
ManufacturerController	Manufacturer operations
ManufacturerPartController	Manufacturer-part operations
SearchController	Consolidated data viewing

Chapter 6

RESULTS & SNAPSHOTS

The completed application provides the main functions implemented in the Product Data Management System. The following screenshots can be used to show the working pages of the application.

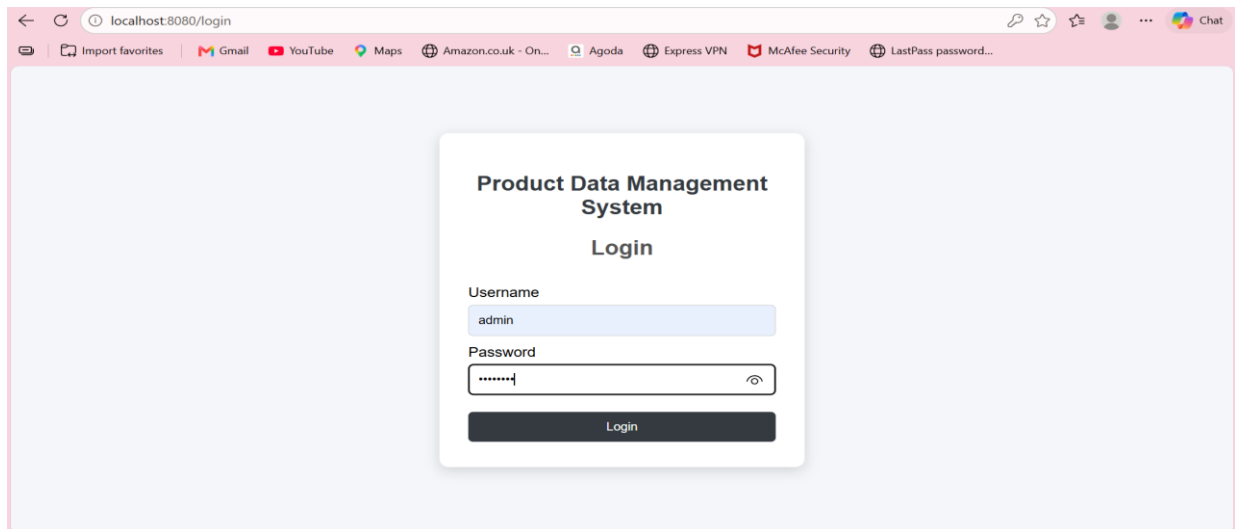


Figure 6.1 – Login page

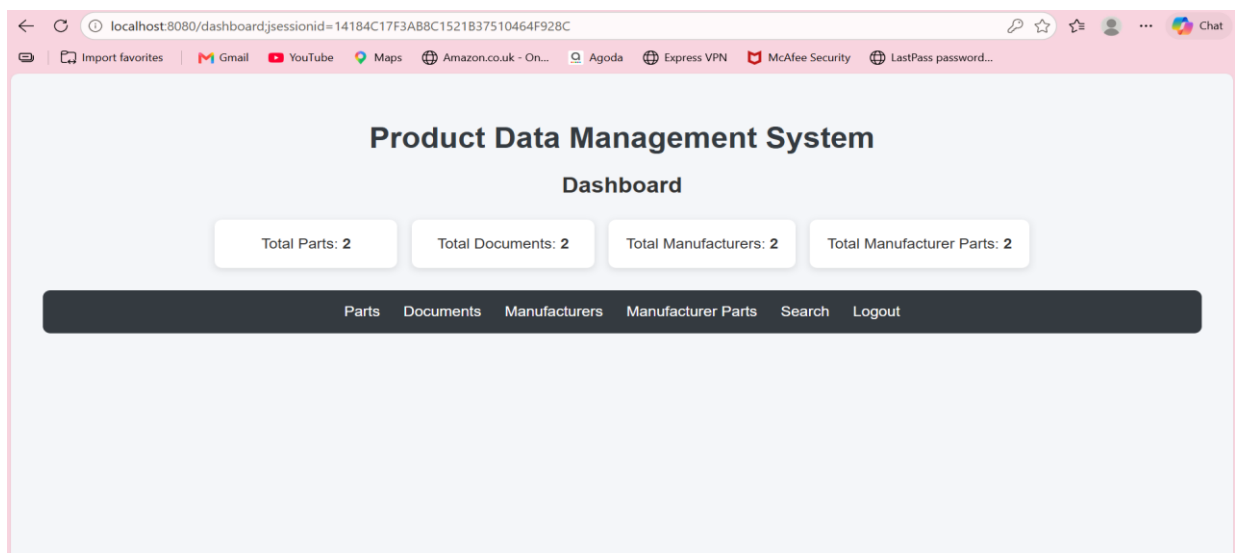


Figure 6.2 – Dashboard page

Parts

Dashboard Parts Documents Manufacturers Manufacturer Parts Search Logout

Create Part

Part Type: Description: Part UOM: Create Part

Part List

Part Number	Part Type	Description	Part UOM	Action
100	Motor	Electric Motor	Mechanical	Delete
101	Gear	Steel gear used in transmission	Mechanical	Delete

Figure 6.3 – Parts page

Documents

Dashboard Parts Documents Manufacturers Manufacturer Parts Search Logout

Create Document

Document Name: Description: Version: Create Document

Document List

Document Number	Name	Description	Version	Action
200	Motor Specification	Technical specification of motor	03	Delete
201	Gear Specification	Steel gear used in transmission	01	Delete

Figure 6.4 – Documents page

Manufacturers

Dashboard Parts Documents Manufacturers Manufacturer Parts Search Logout

Create Manufacturer

Manufacturer Name: Country: Create Manufacturer

Manufacturer List

Manufacturer ID	Name	Country	Action
300	XYZ industries	Canada	Delete
301	ABC Industries	India	Delete

Figure 6.5 – Manufacturers page

Manufacturer Parts

Dashboard Parts Documents Manufacturers Manufacturer Parts Search Logout

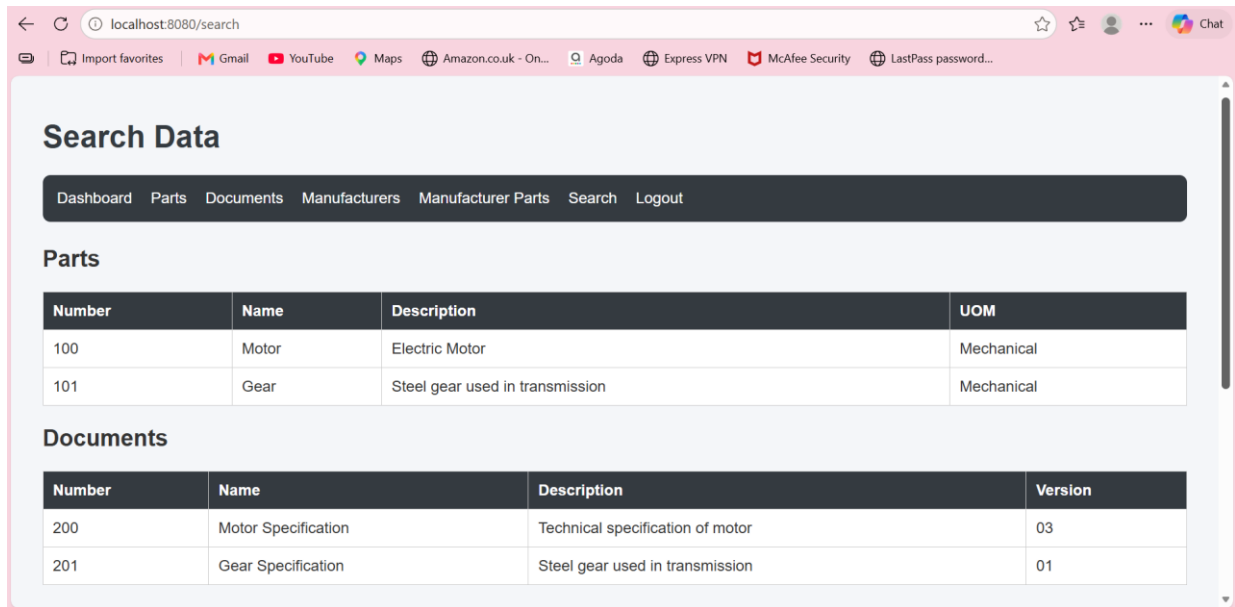
Create Manufacturer Part

Manufacturer Name: Linked Part Number: Create Manufacturer Part

Manufacturer Part List

MFR Part Number	Manufacturer	Linked Part	Action
400	SanjanaTech	100	Delete
401	Srujan	101	Delete

Figure 6.6 – Manufacturer Parts page



Search Data

Dashboard Parts Documents Manufacturers Manufacturer Parts Search Logout

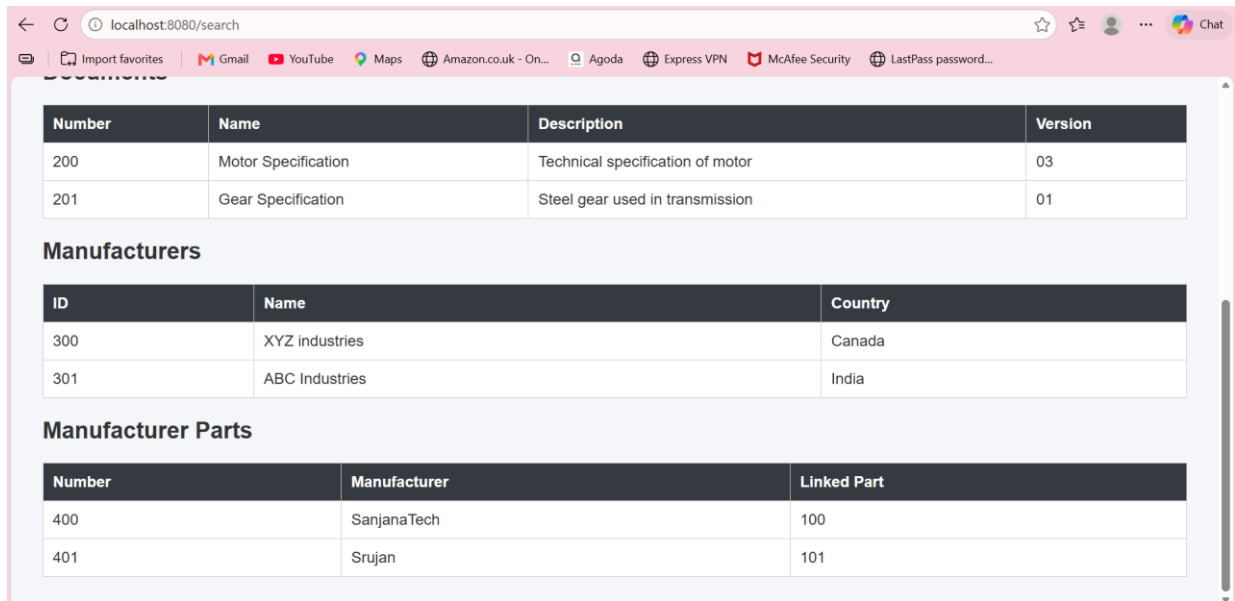
Parts

Number	Name	Description	UOM
100	Motor	Electric Motor	Mechanical
101	Gear	Steel gear used in transmission	Mechanical

Documents

Number	Name	Description	Version
200	Motor Specification	Technical specification of motor	03
201	Gear Specification	Steel gear used in transmission	01

Figure 6.7.1 – Search page



Documents

Number	Name	Description	Version
200	Motor Specification	Technical specification of motor	03
201	Gear Specification	Steel gear used in transmission	01

Manufacturers

ID	Name	Country
300	XYZ Industries	Canada
301	ABC Industries	India

Manufacturer Parts

Number	Manufacturer	Linked Part
400	SanjanaTech	100
401	Srujan	101

Figure 6.7.2 – Search page

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left shows the database structure, with the 'parts' table selected under the 'public' schema. The Query Editor in the center contains the following SQL query:

```
SELECT * FROM public.parts
ORDER BY "number" ASC
```

The Data Output pane shows the results of the query, displaying two rows of data from the 'parts' table:

number	description	name	part_type
100	Electric Motor	Motor	Mechanical
101	Steel gear used in transmissi...	Gear	Mechanical

Below the table, a green message bar indicates: "Successfully run. Total query runtime: 305 msec. 2 rows affected."

Figure 6.8.1 – PostgreSQL Parts table

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left shows the database structure, with the 'documents' table selected under the 'public' schema. The Query Editor in the center contains the following SQL query:

```
SELECT * FROM public.documents
ORDER BY "number" ASC
```

The Data Output pane shows the results of the query, displaying two rows of data from the 'documents' table:

number	description	name	version
200	Technical specification of mo...	Motor Specification	03
201	Steel gear used in transmission	Gear Specification	01

Below the table, a green message bar indicates: "Successfully run. Total query runtime: 305 msec. 2 rows affected."

Figure 6.8.2 – PostgreSQL Documents table

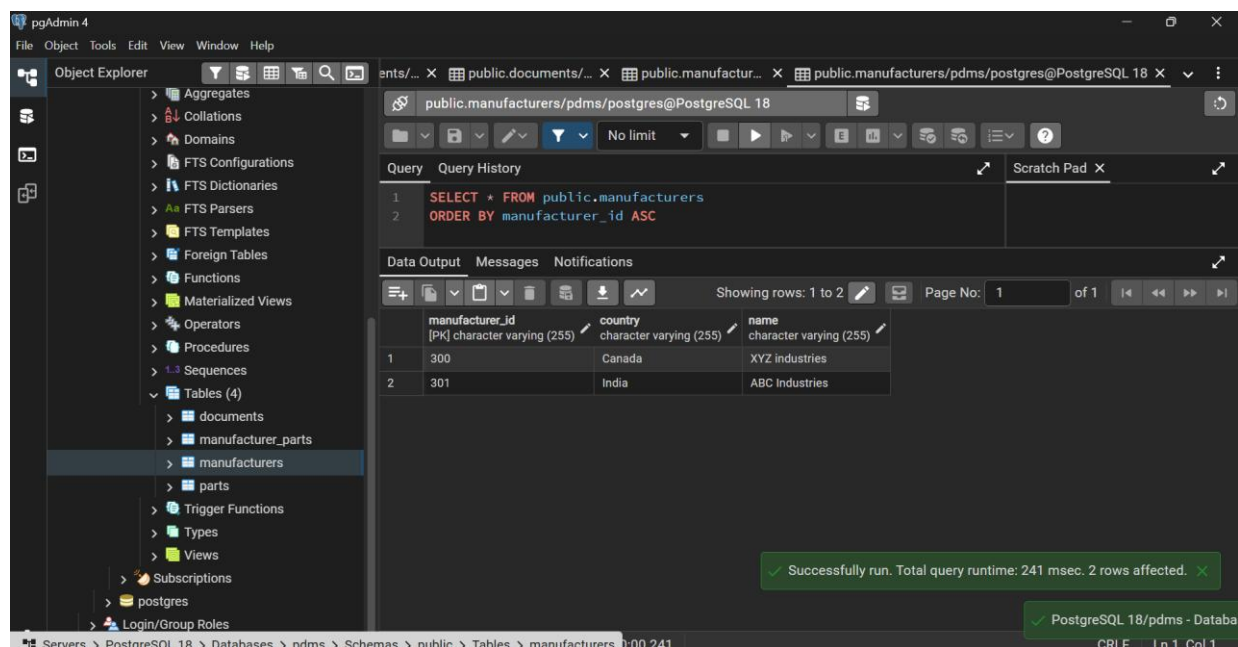


Figure 6.8.3 – PostgreSQL Manufacturer table

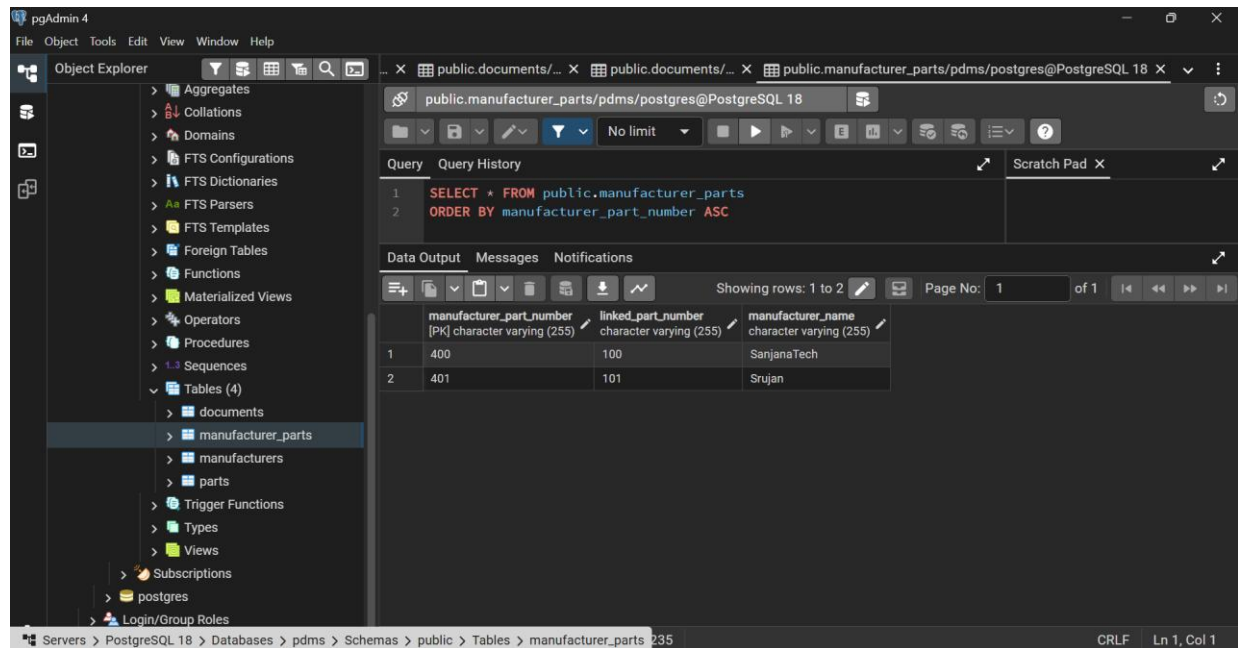


Figure 6.8.4 – PostgreSQL Manufacturer parts table

Chapter 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The Product Data Management System was developed as a simple web application for managing product-related data. The application provides login, dashboard, part management, document management, manufacturer management, manufacturer-part management and consolidated data viewing.

Working on this project during the internship helped me understand how different parts of a Java web application work together. It provided practical experience with Spring Boot, Spring MVC, Thymeleaf, HTML, CSS, Spring Data JPA, Lombok and PostgreSQL, along with practical experience in implementing controllers, services, repositories and database persistence.

7.2 FUTURE ENHANCEMENTS

- Add edit/update operations for the major records.
- Implement real text-based search and filtering.
- Add stronger validation and user-friendly error messages.
- Store users in PostgreSQL instead of using fixed credentials.
- Add secure password hashing and authentication.
- Add Admin and User roles with access control.
- Add charts and richer dashboard information.
- Add report export to PDF or Excel.
- Add automated unit and integration testing.

7.3 LIMITATIONS

- The current login implementation uses fixed credentials in LoginService.
- The current major modules provide create, view and delete operations; edit functionality is not implemented.
- The Search page currently displays consolidated data and does not provide a text-filter form.
- The application depends on a locally configured PostgreSQL database.
- Database credentials are currently configured in application.properties and should be secured for production use.

Appendix

1. SAMPLE DATA

The application generates numeric identifiers automatically for the major records. Part numbers start from 100, document numbers start from 200, manufacturer IDs start from 300, and manufacturer-part numbers start from 400 when new records are created. The actual records depend on the data currently stored in PostgreSQL.

Data Type	Main Fields
Part	Number, Name, Description, Part Type
Document	Number, Name, Description, Version
Manufacturer	Manufacturer ID, Name, Country
Manufacturer Part	Manufacturer Part Number, Manufacturer Name, Linked Part Number
User	Username, Password

2. IMPORTANT PROJECT FILES

File / Folder	Purpose
controller/	Web controllers
service/	Application services
model/	JPA entity/data classes
repository/	Spring Data JPA repositories
templates/	Thymeleaf HTML pages
static/css/	CSS files
application.properties	Spring Boot and PostgreSQL configuration
pom.xml	Maven configuration

3. REFERENCES

- Java documentation
- Spring Boot documentation
- Spring MVC documentation
- Spring Data JPA documentation
- Thymeleaf documentation
- PostgreSQL documentation
- Lombok documentation
- Maven documentation